The upper left part of Figure 4.8 shows a decay chain from fission of ^{235}U that ends in ^{99}Ru, the most stable isobar of $A = 99$. The lower diagram shows that the ^{99}Mo β^- decays all reaches the spin/parity ½-level (see § 3.6.1 and Ch. 11), designated ^{99m}Tc; this isomer decays with $t_{½}$ 6.02 h to long-lived ^{99}Tc, emitting a single γ of 0.142 MeV (> 99%, see upper diagram). The isomer ^{99m}Tc is a widely used radionuclide in nuclear diagnostics (§18.5), and can be conveniently "milked" from its mother ^{99}Mo, see §5.16.

4.9. OTHER MODES OF INSTABILITY

In this chapter we have stressed nuclear instability to beta decay. However, in §4.4 it was learned that very heavy nuclei are unstable to fission. There is also a possibility of instability to emission of α-particles in heavy elements and to neutron and proton emission.

Nuclei are unstable to various forms of decay as indicated in Figure 4.1. For example, making a vertical cut at $N = 100$, the instability from the top is first proton emission, then, α-emission (for $N = 60$ it would instead be positron emission or electron capture, as these two processes are about equally probable), and, after passing the stable nuclides (the isotones ^{170}Yb, ^{169}Tm and ^{168}Er), β^-emission and, finally, neutron emission. This is more clearly indicated in Appendix C, and for the heaviest nuclides (i.e. $Z \geq 81$) in Figures 5.1 and 14.1. For α-decay the Figure indicates that for $A > 150$ ($Z \geq 70$, $N > 80$) the nuclei are α-unstable, but in fact α-decay is commonly observed only above $A \approx 200$. This is due to the necessity for the α-particle to pass over or penetrate the Coulomb barrier (cf. §6.7.3). Although neutron and proton emissions are possible energetically, they are not commonly observed as the competing β-decay processes are much faster.

4.10. EXERCISES

4.1. Calculate the nucleon binding energy in ^{24}Mg from the atomic mass excess value in Table 4.1.

4.2. How many times larger is the nucleon binding energy in ^{24}Na than the electron binding energy when the ionization potential of the sodium atom is 5.14 V?

4.3. Assuming that in the fission of a uranium atom an energy amount of 200 MeV is released, how far would 1 g of ^{235}U drive a car which consumes 1 liter of gasoline (density 0.70 g cm^{-3}) for each 10 km? The combustion heat of octane is 5500 kJ mole^{-1}, and the combustion engine has an efficiency of 18%.

4.4. Estimate if fusion of deuterium into helium releases more or less energy per gram of material consumed than the fission of uranium.

4.5. When a neutron is captured in a nucleus, the mass number of the isotope increases one unit. In the following Table mass excess values are given for three important isotope pairs:

^{235}U	40 915 keV	^{236}U	42 441 keV
^{238}U	47 306	^{239}U	50 571
^{239}Pu	48 585	^{240}Pu	50 122

If the average nucleon binding energy in this region is 7.57 MeV one can calculate the difference between this average binding energy and the one really observed in the formation of ^{236}U, ^{239}U, and ^{240}Pu. Calculate this difference. Discuss the possible significance of the large differences observed for the ^{238}U/^{239}U pair as compared to the other pairs in terms of nuclear power.

4.6. With the semiempirical mass equation (4.15) estimate the binding energy per nucleon for ^{10}B, ^{27}Al, ^{59}Co, and ^{236}U. Compare the results with the observed values in Table 4.1.

4.7. With eqn. (4.17) determine the atomic number corresponding to maximum stability for $A = 10$, 27, 59, and 239. Compare these results with the data in the isotope chart, Appendix C.

4.11. LITERATURE

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